

Digital Forensics Practical Report

File Carving, Hex Analysis, Metadata Examination & TSK Analysis

1. Case Information

Field	Details
Lab Title	File Carving, Metadata, and File System Recovery Lab
Examiner	Fasinu Temidayo
Date of Analysis	2 September 2026 – 4 September 2026
Lab Environment	Kali Linux (working directory ~/ <code>forensic_carving_lab</code>)
Role	Junior Digital Forensic Analyst

Timeline (from `logs/timeline.txt`): Lab started: Wed Sep 2 17:54:13 EDT 2026

2. Evidence Handling

2.1 Evidence Files

Filename	Description	Size
J_ub_law.jpg	JPEG image — primary hex/metadata subject	1,692,150 bytes
120M.7z	Compressed archive containing the USB FAT image	36,720,470 bytes
usb_fat_carving.001 (inside 120M/)	Raw USB disk image (FAT), source for Scalpel carving	124,780,544 bytes
Ch01InChap01.dd	Raw FAT12 floppy/USB image, source for TSK analysis	1,474,560 bytes

2.2 Cryptographic Hashes (from `evidence_hashes.txt`)

Recorded: Wed Sep 2 18:04:22 EDT 2026

File	MD5	SHA-256
J_ub_law.jpg	83a360ac7f7e0ca318a50ff639395f509	2850f639395f509752c0e8b14fcff8ec7dc29e23914dbc435f8ef99

File	MD5	SHA-256
120M.7z	06c0e8b720827909d439308979c882641349341d5527f9d2cdb979b8c67fc86c5abe7b33f	495308979c882641349341d5527f9d2cdb979b8c67fc86c5abe7b33f
Ch01InChap01.dd	a117773bcf1fc88ec0ab9805349f31918ab98b3aadb2480685efb8e4980d34297b83bd5	ba4a1d0ba49f4a6667b00a3b3e85e604

Screenshot: initial hashes — `screenshots/01_evidence.png`, `screenshots/03_hashes.png`

2.3 Original Acquisition Record for the USB Image

The USB image `usb_fat_carving.001` was originally acquired with FTK Imager (metadata preserved in `120M/usb_fat_carving.001.txt`):

Field	Value
Acquired using	AccessData FTK Imager 4.5.0.3
Case Number	USB File Carving
Evidence Number	UB-002
Examiner (original acquisition)	Frank Xu, University of Baltimore
Source	General UDisk USB Device, 119 MB, 243,712 sectors × 512 bytes
Acquisition MD5	ba4a1d0ba49f4a6667b00a3b3e85e604 (verified)
Acquisition SHA-1	bcc2d49fd49c9521ecb1739f6542c6bf327375ef (verified)

This is the chain-of-custody record from the original imaging of the physical USB drive, prior to it being packaged as `120M.7z` for this lab.

2.4 Evidence Integrity Statement

The original evidence files were not modified during this investigation. All carving and recovery operations were performed against copies, and integrity was checked using MD5/SHA-256 hashing before and after analysis.

3. Part A — File Signatures and Hex Analysis (`J_ub_law.jpg`)

3.1 Header Inspection (`xxd`)

Command:

```
xxd J_ub_law.jpg | head -20
xxd -l 64 J_ub_law.jpg
xxd -p J_ub_law.jpg | head -c 20
```

Result: The file begins with `ff d8 ff e1 17 f0 45 78 69 66 00 00 49 49 2a 00 08 00 00 00` — i.e. `FFD8` (JPEG **SOI** marker) followed by `FFE1` (an **APP1** segment, not **APP0/JFIF**), then the ASCII bytes `45 78 69 66 00 00` = `"Exif\0\0"`, confirming this JPEG carries an **EXIF** header rather than a plain JFIF header.

Screenshot: `screenshots/04_xxd_inspection.png`

3.2 Footer Identification (EOI marker)

Command:

```
xxd -p J_ub_low.jpg | grep -o "ffd9" | wc -l
xxd J_ub_low.jpg | tail -10
```

Analysis: The JPEG **EOI** marker `FFD9` terminates the file. File size confirmed at 1,692,150 bytes via `stat -c%s`.

Screenshot: `screenshots/05_jpeg_footer.png`

3.3 Hex Dumps

Command:

```
xxd -p J_ub_low.jpg > J_ub_low_hexdump.txt
xxd J_ub_low.jpg > J_ub_low_hexdump_formatted.txt
```

Both plain (`-p`, no addresses) and formatted (with addresses/ASCII column) hex dumps were generated successfully.

Screenshot: `screenshots/06_plain_hex.png`, `screenshots/06_plain_hex_000.png`

3.4 Reverse Hexdump Reconstruction

Command:

```
xxd -r -p J_ub_low_hexdump.txt J_ub_low_reconstructed.jpg
xxd -r J_ub_low_hexdump_formatted.txt J_ub_low_reconstructed2.jpg
```

```
file J_ub_low_reconstructed.jpg
file J_ub_low_reconstructed2.jpg
```

```
md5sum J_ub_low.jpg J_ub_low_reconstructed.jpg J_ub_low_reconstructed2.jpg
sha256sum J_ub_low.jpg J_ub_low_reconstructed.jpg J_ub_low_reconstructed2.jpg
```

Result:

File	MD5
J_ub_low.jpg	83a360ac7f7e0ca318e5bfe39f95f137
J_ub_low_reconstructed.jpg	83a360ac7f7e0ca318e5bfe39f95f137

File	MD5
J_ub_law_reconstructed2.jpg	83a360ac7f7e0ca318e5bfe39f95f137

All three MD5 (and matching SHA-256) hashes are **identical**. `file` confirms all three are valid JPEG image data with matching EXIF headers and dimensions. This proves that both the plain (`-p`) and address-formatted hex dumps are lossless, reversible representations of the original binary — reconstructing from either produces a byte-for-byte identical file.

Screenshot: screenshots/07_reverse_dump.png

3.5 JPEG Structure Summary

Element	Value
Header (SOI + marker)	FF D8 FF E1 (APP1/Exif segment)
Footer (EOI)	FF D9
File size	1,692,150 bytes
Reported dimensions	1920×1080 (baseline, 8-bit precision, 3 components)

Screenshot: screenshots/08_Jpeg_structure.png

4. Part B — Embedded Metadata Examination (J_ub_law.jpg)

4.1 Strings Extraction

Command:

```
strings J_ub_law.jpg > jpeg_strings.txt
strings J_ub_law.jpg | grep -i "camera\\|model\\|date\\|copyright\\|howard\\|korn"
```

Readable text extracted from the file's embedded XMP/IPTC metadata block confirmed the presence of camera make/model strings (NIKON CORPORATION, NIKON D4), software strings (Adobe Photoshop 21.1 (Macintosh)), and the artist/copyright name (HOWARD KORN) directly in the raw byte stream — well before running a dedicated EXIF parser.

Screenshot: screenshots/09_embedded_metadata_a, screenshots/09_embedded_metadata_b

4.2 exiftool Extraction

Command:

```
exiftool J_ub_law.jpg > jpeg_exif.txt
exiftool J_ub_law.jpg | grep -E "Camera|Model|Date|Software|Artist"
```

Key metadata fields recovered:

Field	Value
Make	NIKON CORPORATION
Camera Model Name	NIKON D4
Lens Model	17.0-35.0 mm f/2.8
Serial Number	2027750
Focal Length	20.0 mm
F Number	11.0
Exposure Time	8 sec
Exposure Program	Manual
ISO	200
Create Date	2013:10:08 18:56:21
Modify Date	2020:08:20 10:20:05
Software	Adobe Photoshop 21.1 (Macintosh)
Artist / Copyright	HOWARD KORN
Copyright Notice (embedded)	Copyright 1999 Adobe Systems Incorporated
Image Size	1920×1080
Subjects (XMP)	2013, campus, Merrick School of Business
Hierarchical Subject	Schools Merrick School of Business

GPS coordinates and lens-profile-specific distortion data were **not** present in the metadata.

Screenshot: `screenshots/10_metadata.png`

4.3 Findings Summary

- **Original capture date:** 8 October 2013, 18:56:21 (Nikon D4, manual exposure, 20mm/f11/8s/ISO200)
- **Last edited:** 20 August 2020, 10:20:05, via Adobe Photoshop 21.1 on macOS
- **Editing history (from embedded XMP `xmpMM:History`):** the file went through multiple Photoshop/Camera Raw save-and-convert cycles between January 2014 and August 2020 (RAW→TIFF→JPEG conversions, multiple re-saves), indicating repeated post-processing over several years rather than a single edit.
- **Attribution:** Artist/Copyright field names “HOWARD KORN”
- **Content clue:** XMP subject tags tie the image to “Merrick School of Business” and the year 2013
- **Not found:** GPS coordinates, lens profile/distortion correction data

Screenshot: screenshots/11_findings.png

Note on the camera-model finding: an initial pass using `strings/grep` only surfaced the string `NIKON CORPORATION` for the “Make” field. Running `exiftool` (a full EXIF/XMP parser rather than a raw string search) additionally recovered the specific **Camera Model Name: NIKON D4** and lens details, which raw string grepping missed because they were stored in structured binary EXIF tags rather than as standalone ASCII strings.

5. Part C — The Sleuth Kit (TSK) Analysis of Ch01InChap01.dd

5.1 Image Information

Command:

```
img_stat Ch01InChap01.dd
```

Output (tsk_analysis/img_stat.txt):

```
IMAGE FILE INFORMATION
```

```
-----  
Image Type: raw
```

```
Size in bytes: 1474560
```

```
Sector size: 512
```

5.2 Volume System Analysis

Command:

```
mmls Ch01InChap01.dd
```

Output (tsk_analysis/mmls.txt): empty / no output.

Analysis: `mmls` returned no partition table, and this is corroborated by `tsk_summary.txt`, which records “**Partition scheme: 0 partitions.**” This means `Ch01InChap01.dd` is a **non-partitioned** image — a raw FAT12 file system written directly to the media (typical of a floppy disk), with no MBR/partition table layer to parse. This is why `fsstat/fls` below work directly against the image with no `-o` offset needed.

Screenshot: screenshots/12_sleuthkit.png

5.3 File System Information

Command:

```
fsstat Ch01InChap01.dd
```

Output (tsk_analysis/fsstat.txt):

FILE SYSTEM INFORMATION

File System Type: FAT12

OEM Name: kOnIHC

Volume ID: 0x0

Sectors before file system: 0

File System Layout (in sectors)

Total Range: 0 - 2879

* Reserved: 0 - 0

** Boot Sector: 0

* FAT 0: 1 - 9

* FAT 1: 10 - 18

* Data Area: 19 - 2879

** Root Directory: 19 - 32

** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782

Root Directory: 2

CONTENT INFORMATION

Sector Size: 512

Cluster Size: 512

Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)

33-236 (204) -> EOF

285-311 (27) -> EOF

Analysis: Confirms a FAT12 file system (consistent with a small floppy/USB volume), 512-byte sectors and clusters, a 9-sector FAT table (×2 copies), and a root directory spanning sectors 19–32. The FAT contents show two allocated cluster runs terminating in EOF — sectors 33–236 and 285–311 — which map to the two allocated files identified below (*Client Info.mdb* and *Income.xls*).

Screenshot: `screenshots/13_analysis.png`

5.4 List All Files

Command:

```
fls Ch01InChap01.dd
```

Output (tsk_analysis/fls_output.txt):

```
r/r 5:      Client Info.mdb
r/r * 8:    Billing Letter.doc
r/r * 11:   confirmation.txt
r/r 13:     Income.xls
r/r * 15:   letter1.txt
r/r * 17:   Regrets.doc
v/v 45779:  $MBR
v/v 45780:  $FAT1
v/v 45781:  $FAT2
V/V 45782:  $OrphanFiles
```

Analysis: 6 user files total — 2 allocated (Client Info.mdb, Income.xls), 4 deleted (marked with *: Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc). The v/v / V/V entries are TSK's virtual metadata objects (not real files on the volume).

Screenshot: screenshots/14_TSK_findings.png

Note: The `fls -i inode Ch01InChap01.dd` command produced no output (`tsk_analysis/fls_inode.txt` was empty) — `-i` is TSK's flag for specifying *image type* (e.g. `-i raw`), not a way to list inode numbers, so `-i inode` was not a valid image type and the command did not run as intended. Inode numbers are already shown by default in the plain `fls` output above (the numbers after `r/r`).

5.5 Deleted File Recovery

Cross-referencing the deleted entries from `fls` against inode numbers gives:

Inode	Filename	Status
8	Billing Letter.doc	Deleted
11	confirmation.txt	Deleted
13	Income.xls	Allocated
15	letter1.txt	Deleted
17	Regrets.doc	Deleted

Recovery commands used:

```
icat Ch01InChap01.dd 13 > recovered_files/Income.xls
```


6. Part D — Embedded File Discovery with Binwalk (J_ub_law.jpg)

Command:

```
binwalk J_ub_law.jpg > binwalk_jpeg_scan.txt
binwalk -v J_ub_law.jpg >> binwalk_jpeg_scan.txt
```

Output (binwalk_jpeg_scan.txt):

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

```
Scan Time:      2026-09-04 15:00:28
Target File:    /home/kali/forensic_carving_lab/J_ub_law.jpg
MD5 Checksum:  83a360ac7f7e0ca318e5bfe39f95f137
Signatures:    436
```

Analysis: Binwalk confirms the file is a single, well-formed JPEG with an EXIF/TIFF metadata block starting at offset 0xC, and independently surfaces the embedded copyright string found earlier via `strings`. No secondary embedded file (e.g. an appended ZIP or second image signature) was detected — the scan found no additional file signatures beyond the JPEG/EXIF/TIFF structure itself, i.e. nothing was hidden after the JPEG's EOI marker.

Screenshot: `screenshots/15_binwalk_scan`, `screenshots/16_binwalk_scan`

7. Part E — USB Image Preparation and Analysis

7.1 Extraction

Command:

```
7z x 120M.7z
cd 120M
ls -la
```

Result: Extraction produced `120M/usb_fat_carving.001` (124,780,544 bytes), accompanied by the original FTK Imager acquisition record `usb_fat_carving.001.txt` (see Section 2.3).

Screenshot: `screenshots/17_extracting_usb.png`, `screenshots/18_extracting_usb_b.png`

7.2 TSK Analysis of the USB Image

Commands:

```
md5sum usb_fat_carving.001 >> ../evidence_hashes.txt
sha256sum usb_fat_carving.001 >> ../evidence_hashes.txt
img_stat usb_fat_carving.001 > ../tsk_analysis/usb_img_stat.txt
mmls usb_fat_carving.001 > ../tsk_analysis/usb_mmls.txt
fsstat -o 128 usb_fat_carving.001 > ../tsk_analysis/usb_fsstat.txt
```

Note: the text-file redirects for this step (`usb_img_stat.txt`, `usb_mmls.txt`, `usb_fsstat.txt`) were not present in the submitted lab folder — only the on-screen results were captured, via `screenshots/20_fat_carving1.png` and `screenshots/20_fat_carving2.png`. If you still have terminal scrollback or re-run these three commands, paste the actual `mmls` partition table and `fsstat -o 128` output here so the offset (partition start sector, typically 128 for this FAT image) is documented from your own run rather than assumed.

Screenshot: `screenshots/20_fat_carving1.png`, `screenshots/20_fat_carving2.png`

8. Part F — Carving Deleted Files with Scalpel (`usb_fat_carving.001`)

8.1 Configuration

`scalpel.conf` was edited to enable JPEG carving by uncommenting three signature rules:

```
jpg    y    5000000    \xff\xd8\xff\xe0\x00\x10\x4a\x46    \xff\xd9
jpg    y    5000000    \xff\xd8\xff\xe1\x00\x10\x45\x78    \xff\xd9
jpg    y    5000000    \xff\xd8\xff\xdb                                \xff\xd9
```

These match JFIF (APP0), Exif (APP1), and generic quantization-table-led JPEGs respectively, each bounded by the FFD9 EOI footer, with a 5,000,000-byte max carve size.

Screenshot: `screenshots/21_scalpel.png`

8.2 Carving Run

Command:

```
scalpel -c scalpel.conf -o scalpel_output usb_fat_carving.001
```

Result (`scalpel_output/audit.txt`): Scalpel carved **10 JPEG** files from the raw USB image:

File	Start Offset	Length (bytes)
00000000.jpg	335,872	85,046
00000001.jpg	425,822	85,046
00000002.jpg	3,796,992	99,347
00000003.jpg	35,350,242	239,079

File	Start Offset	Length (bytes)
00000004.jpg	35,590,996	13,715
00000005.jpg	36,352,448	2,946
00000006.jpg	36,393,610	2,268
00000007.jpg	36,571,136	99,277
00000008.jpg	36,671,488	292,225
00000009.jpg	35,903,812	36,643

scalpel_output/carving_summary.txt confirms: **10 jpg, 2 txt** files recovered in total.

Screenshot: screenshots/22_scalpel_results

8.3 Review of Carved JPEGs

Command:

```
for file in *.jpg; do file $file; md5sum $file; done > jpeg_details.txt
```

Findings (jpg-0-0/jpeg_details.txt):

File	Dimensions	Notes
00000000.jpg / 00000001.jpg	360×540	Identical content — same MD5 (597cef9f...), duplicate carve
00000002.jpg	618×927	Progressive JPEG
00000003.jpg	1667×762	Comment: “Software: Snipaste” — a screenshot tool
00000004.jpg	604×158	Comment: “Software: Snipaste”
00000005.jpg / 00000006.jpg	256×144	Small thumbnails
00000007.jpg / 00000008.jpg	750×1334	Progressive JPEG, phone-screen aspect ratio
00000009.jpg (in jpg-2-0/)	—	Carved separately due to a different signature match

Screenshot: screenshots/23_scalpel_img1.png, screenshots/24_scalpel_file_review.png, screenshots/25_scalpel_image1.png

8.4 Hash Verification of Recovered JPEGs

Command:

```
for file in *.jpg; do
  md5sum "$file" >> recovered_hashes.txt
  sha256sum "$file" >> recovered_hashes.txt
done
```

Result (recovered_hashes.txt):

File	MD5	Size
00000000.jpg	597cef9fa6bab3b73312e8e274364b04	85,046 B
00000001.jpg	597cef9fa6bab3b73312e8e274364b04	85,046 B
00000002.jpg	e833e245176b08a46508e26961116bb4	99,347 B
00000003.jpg	9d09c214f6fa49ad4b39fff15e22af04	239,079 B
00000004.jpg	3dcb6a4ca242b85d904da775fcd8a4b1	13,715 B
00000005.jpg	d09b9174b6bc9b09bcf84724509066a2	2,946 B
00000006.jpg	3e1ad4207d62d924fb6bc5473d842e13	2,268 B
00000007.jpg	a4fc766ba1a38f36263d0bfe6fc0ac1d	99,277 B
00000008.jpg	ca42511022040f8739f8b2a8445c81b6	292,225 B

Analysis: All recovered files hash-verify as valid, complete JPEGs with matching file type identification — Scalpel’s header/footer carving successfully reconstructed full, viewable images from unallocated space on the USB image, with one exact duplicate pair (00000000/00000001) carved from adjacent, overlapping regions.

Screenshot: screenshots/26_recovered_hashes.png

9. Answers to Key Questions

Q1: What JPEG header/footer signatures were identified in J_ub_law.jpg, and were the hex dump and its reversal identical to the original?

A: Header: FF D8 FF E1 + "Exif\0\0" (APP1/EXIF-tagged JPEG, not plain JFIF). Footer: FF D9 (EOI). Both the plain (xxd -p) and formatted (xxd) hex dumps, when reversed with xxd -r, reproduced files with MD5 83a360ac7f7e0ca318e5bfe39f95f137 — **identical** to the original J_ub_law.jpg, confirming lossless reversibility.

Q2: What metadata was recoverable from J_ub_law.jpg, and by which method?

A: strings alone surfaced the make (NIKON CORPORATION), software (Adobe Photoshop 21.1), and artist (HOWARD KORN) as raw ASCII. exiftool additionally decoded structured EXIF/XMP fields not visible to plain string search — camera model (NIKON D4), lens (17-35mm f/2.8), exposure settings (f/11, 8s, ISO 200, manual), original capture date (8 Oct 2013) vs. last-modified date (20 Aug 2020), and XMP subject tags (“Merrick School of Business”, 2013). No GPS data was present.

Q3: How many deleted files exist in Ch01InChap01.dd, and how was recovery verified?

A: 4 deleted files (Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc) per fls output, alongside 2 allocated files (Client Info.mdb, Income.xls). Income.xls (inode 13) was recovered via icat.

Q4: How many files did Scalpel carve from the USB image, and were they valid?

A: 10 JPEG files + 2 text files. All 9 unique JPEGs (00000000–00000008, with 00000000/00000001 being duplicates) verified as structurally valid JPEG image data via file, with individually recorded MD5/SHA-256 hashes for chain-of-custody.

10. Summary of Findings

Item	Finding
JPEG evidence file	J_ub_law.jpg, 1,692,150 bytes, EXIF-tagged JPEG
Hex reconstruction	Verified lossless (identical MD5/SHA-256 to original)
Camera/metadata identified	Nikon D4, HOWARD KORN (copyright), edited in Photoshop 21.1, captured 2013, last modified 2020
Binwalk result	Single JPEG structure, no additional embedded/hidden files detected
Ch01InChap01.dd file system	FAT12, no partition table (raw floppy layout)
Ch01InChap01.dd files	6 total — 2 allocated, 4 deleted
USB image (usb_fat_carving.001)	124,780,544 bytes, FTK-acquired, FAT
Scalpel carving result	10 JPEG + 2 TXT files recovered from unallocated space

Item	Finding
Evidence integrity	Verified via MD5/SHA-256 throughout

11. Conclusion

This lab exercised the full forensic file-recovery pipeline: low-level hex/signature analysis and lossless reconstruction of a JPEG, embedded-metadata extraction (both raw string search and structured EXIF/XMP parsing), TSK-based file system and deleted-file analysis of a FAT12 image, binwalk-based embedded-file discovery, and signature-based file carving of unallocated space on a larger USB image using Scalpel. All recovered artifacts were hash-verified, and no modification was made to the original evidence files at any stage.

12. Appendices

Appendix A: Files Present in Submission

- evidence_hashes.txt, jpeg_analysis.txt, metadata_analysis.txt, jpeg_metadata.txt, jpeg_exif.txt, jpeg_strings.txt
- J_ub_low_hexdump.txt, J_ub_low_hexdump_formatted.txt
- binwalk_jpeg_scan.txt
- tsk_analysis/ — img_stat.txt, mmls.txt (empty), fsstat.txt, fls_output.txt, fls_inode.txt (empty), tsk_summary.txt
- scalpel.conf, scalpel_output/audit.txt, scalpel_output/carving_summary.txt, scalpel_output/jpg-0-0/, scalpel_output/jpg-2-0/
- 120M/usb_fat_carving.001, 120M/usb_fat_carving.001.txt

Appendix B: Recovered/Reconstructed Files

- J_ub_low_reconstructed.jpg, J_ub_low_reconstructed2.jpg (both hash-identical to original)
- 9 unique carved JPEGs (00000000.jpg-00000008.jpg) from usb_fat_carving.001
- Income.xls recovered from Ch01InChap01.dd via icat (inode 13)

Appendix C: Screenshots

All screenshots are in the /screenshots folder (26 files, 01_evidence.png through 26_recovered_hashes.png).

Appendix D: Tools Used

- xxd, strings, exiftool, binwalk
- The Sleuth Kit: img_stat, mmls, fsstat, fls, icat

- 7z, scalpel
 - md5sum, sha256sum, file
-

13. Evidence Integrity Declaration

I hereby certify that:

- The original evidence files were not modified during this investigation
- All analysis was performed using validated forensic tools
- All recovered evidence was verified using cryptographic hashing
- Chain of custody was maintained throughout the investigation

Signature: Fasinu Temidayo Lucky

Date: 4th September 2026